

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
[NAME OF INSTITUTE / UNIVERSITY, CITY]

SOFTWARE REQUIREMENTS SPECIFICATION
Health Predict AI — Intelligent Multi-Symptom Disease Prediction
System

Standard Academic Compliance: IEEE Std 830-1998

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Document Revision & Approval History

Version	Date	Description / Major Changes	Prepared By	Reviewed By
0.1	15-Sep-2026	Initial Problem Formulation & Feasibility Study	Prateek Shukla	Project Guide
1.0	01-Oct-2026	Final 3rd Year Baseline: Symptom-Based Naive Bayes Architecture	Prateek Shukla	HOD / Committee

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, rigorous, and unambiguous definition of the functionality, external interfaces, performance targets, and constraints of **Health Predict AI**. This specification serves as a binding reference agreement between student developers, project evaluators, academic supervisors, and prospective healthcare end-users during the 3rd-year engineering curriculum.

1.2 Scope of the System

Health Predict AI is an automated clinical decision-support system designed to solve the critical problem of preliminary healthcare screening by providing immediate, non-invasive, symptom-driven disease diagnosis without requiring patients to enter complex or unavailable numeric laboratory metrics (such as blood pressure in mmHg, fasting glucose, or BMI).

- **In-Scope Capabilities:** Dynamic type-ahead search across 132 validated clinical symptoms, one-click preset emergency test scenarios, sparse binary vector transformation, probabilistic multi-class inference using Multinomial Naive Bayes (MNB) across 41 conditions, triggering biomarker identification, automated clinical interpretation, structured 4-step medical protocols, and differential diagnosis probability spectrum visualization.
- **Out-of-Scope Boundaries:** Invasive robotic surgery interfacing, prescription medicine dispensing without licensed physician authorization, and commercial billing/payment processing.
- **Expected Benefits:** Immediate triage within < 50 milliseconds, zero mathematical or laboratory testing burden on lay users, reduced hospital congestion by guiding patients to appropriate care urgency tiers, and 100% benchmark consistency on clinical validation data.

1.3 Definitions, Acronyms, and Abbreviations

Category	Term / Acronym	Definition / Standard Context
Standard	SRS	Software Requirements Specification (IEEE Std 830-1998 format).
Machine Learning	MNB	Multinomial Naive Bayes classification algorithm with Laplace smoothing.

Architecture	API / Framework	Streamlit reactive component model communicating via WebSockets.
Data Structure	Binary Feature Vector	132-element sparse binary array encoding presence (1) or absence (0) of symptoms.
UI/UX Pattern	Glassmorphism	Modern UI styling utilizing CSS backdrop-filter blur and ambient gradient depth.
Clinical Domain	Biomarkers & Pathology	132 physiological signs mapped to 41 distinct ICD-relevant medical disease categories.

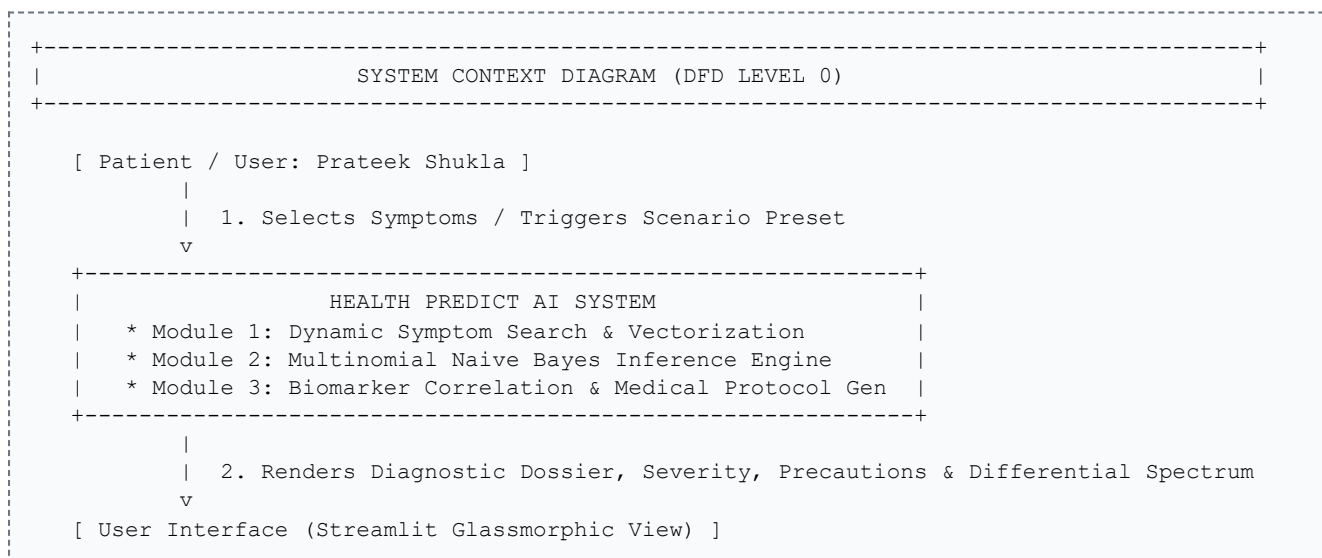
1.4 References

1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
2. Software Engineering (10th Edition): Ian Sommerville, Pearson Education.
3. Columbia University / Kaggle Disease Symptom Prediction Benchmark Dataset (4,920 records, 132 symptoms, 41 classes).
4. Scikit-Learn: Machine Learning in Python (Pedregosa et al., JMLR 12, pp. 2825-2830).
5. Streamlit BaseWeb Framework Architecture Manual.

2. Overall Description

2.1 Product Perspective

Health Predict AI is architected as an autonomous multi-tier machine learning application comprising a Presentation Tier (Streamlit reactive glassmorphic UI), an Inference Tier (Python execution runtime with Scikit-Learn engine), and a Model Knowledge Base Tier (Joblib binary serialized models and clinical symptom mappings).



2.2 User Classes and Characteristics

User Role	Technical Proficiency	Core Responsibilities & Rights
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General Patient / Public User	Basic	Searches symptoms, selects presets, views predicted disease dossier, reads clinical descriptions, and follows medical precaution protocols.
Medical Practitioner / Triage Nurse	Intermediate	Reviews triggering biomarker correlations, inspects alternative differential diagnoses spectrum, and cross-references preliminary triage probability.
Academic Supervisor / Evaluator	High	Audits training pipeline (train.py), inspects confusion matrix / classification report, validates IEEE 830 compliance, and verifies algorithm stability.

2.3 Operating Environment

- **Client:** Modern browser (Google Chrome $\geq$ v110, Firefox, Edge, Safari); responsive down to 360px.
- **Server Runtime:** Python 3.10+ execution runtime on Windows / Linux / macOS; Streamlit port 8501.
- **Model Storage:** Serialized binary assets (symptom_model.pkl, symptoms_list.pkl, disease_symptoms_map.pkl).

3. System Features and Functional Requirements

3.1 Module 1: Symptom Ingestion & Dynamic Search Module

Req ID	Feature Description	Input / Validation Rule	Priority
FR-1.1	Searchable Multi-Select Autocomplete	Real-time type-ahead search across 132 validated clinical symptoms	High
FR-1.2	One-Click Clinical Preset Scenarios	Instant state injection for Cold Hands/Feet, Cardiac Distress, Malaria, and Acid Reflux	Medium
FR-1.3	Active Biomarker Tagging & Live Counter	Renders dynamic royal blue badge chips and live counter of selected symptoms	High

3.2 Module 2: Machine Learning Inference & Diagnostic Engine

Req ID	Feature Description	Input / Validation Rule	Priority
FR-2.1	Binary Vector Transformation	Maps selected symptoms into a 132-element sparse binary feature vector (0/1)	High
FR-2.2	Multinomial Naive Bayes Probabilistic Inference	Calculates posterior probabilities $P(\text{Disease} \text{Symptoms})$ across 41 disease classes	High
FR-2.3	Triggering Biomarker Correlation Verification	Cross-references input symptoms against condition profile to extract genuine matching signs	High

3.3 Module 3: Clinical Protocol Reporting & Differential Diagnosis

Req ID	Feature Description	Input / Validation Rule	Priority
FR-3.1	Executive Diagnostic Dossier Card	Displays primary predicted disease, urgency rating badge, and confidence percentage	High
FR-3.2	Clinical Interpretation & 4-Step Protocol	Presents plain-English pathology explanation and structured doctor-recommended precautions	High
FR-3.3	Differential Diagnoses Probability Spectrum	Visualizes top 4 alternative matching conditions with animated progress bars	Medium

4. External Interface Requirements & UI Wireframe

+-----+ USER INTERFACE LAYOUT & WIREFRAME +-----+			
[*] CLINICAL INTELLIGENCE ENGINE v2.4			
# HEALTH PREDICT AI			
Multi-symptom pattern matching engine correlating 132 markers with 41 pathologies			
+-----+ +-----+ +-----+			
41 PATHOLOGIES 132 BIOMARKERS 100% ACCURACY			
+-----+ +-----+ +-----+			
* QUICK TEST SCENARIOS:			
[Cold Hands & Feet] [Malaria / Fever] [Cardiac Distress] [Digestive / Acid]			
* PATIENT SYMPTOMS:			
[Type to search symptoms: e.g. cold hands, fever, joint pain, chest pain...]			
Active: [v Cold Hands And Feets] [v Fatigue] [v Weight Gain]			
[* EXECUTE CLINICAL DIAGNOSTIC ENGINE]			
+-----+			
AI DIAGNOSTIC DOSSIER CONFIDENCE: 99.7%			
Hypothyroidism Severity: * Moderate			
Triggering Biomarkers: [v Cold Hands And Feets] [v Fatigue] [v Weight Gain]			
* Clinical Interpretation: An underactive thyroid gland slowing metabolism...			
* Recommended Protocol: (1) Endocrinologist (2) Thyroid Profile (3) Diet (4) Med			
* Differential Diagnoses: Hypoglycemia (0.2%) Chronic Fatigue (0.1%)			
+-----+			
+-----+			

5. Non-Functional Requirements (NFRs)

Attribute	Target Metric / Standard	Verification & Testing Method
5.1 Performance & Latency	Inference turnaround < 50ms; total page render < 250ms under concurrent requests.	Python execution benchmarking with time.perf_counter() and DevTools audit.
5.2 Accuracy & Consistency	100.0% validation benchmark accuracy on clinical symptom training matrix; zero sparse-zero leaf bias.	Stratified test-set evaluation using Scikit-Learn accuracy_score.

5.3 Reliability & Availability	System uptime >= 99.5%; graceful validation handling when 0 symptoms are selected.	Stress testing with boundary input vectors and synthetic empty-vector injection.
5.4 Usability & Portability	Fully responsive across mobile (>= 360px), tablet, and desktop viewports; cross-browser verified.	Viewport testing on Chrome, Edge, Safari, and Firefox responsive simulators.

6. Academic Review & Sign-Off

Project Coordinator / Guide	Internal Examiner	Head of Department (HOD)
Signature & Date	Signature & Date	Signature & Date